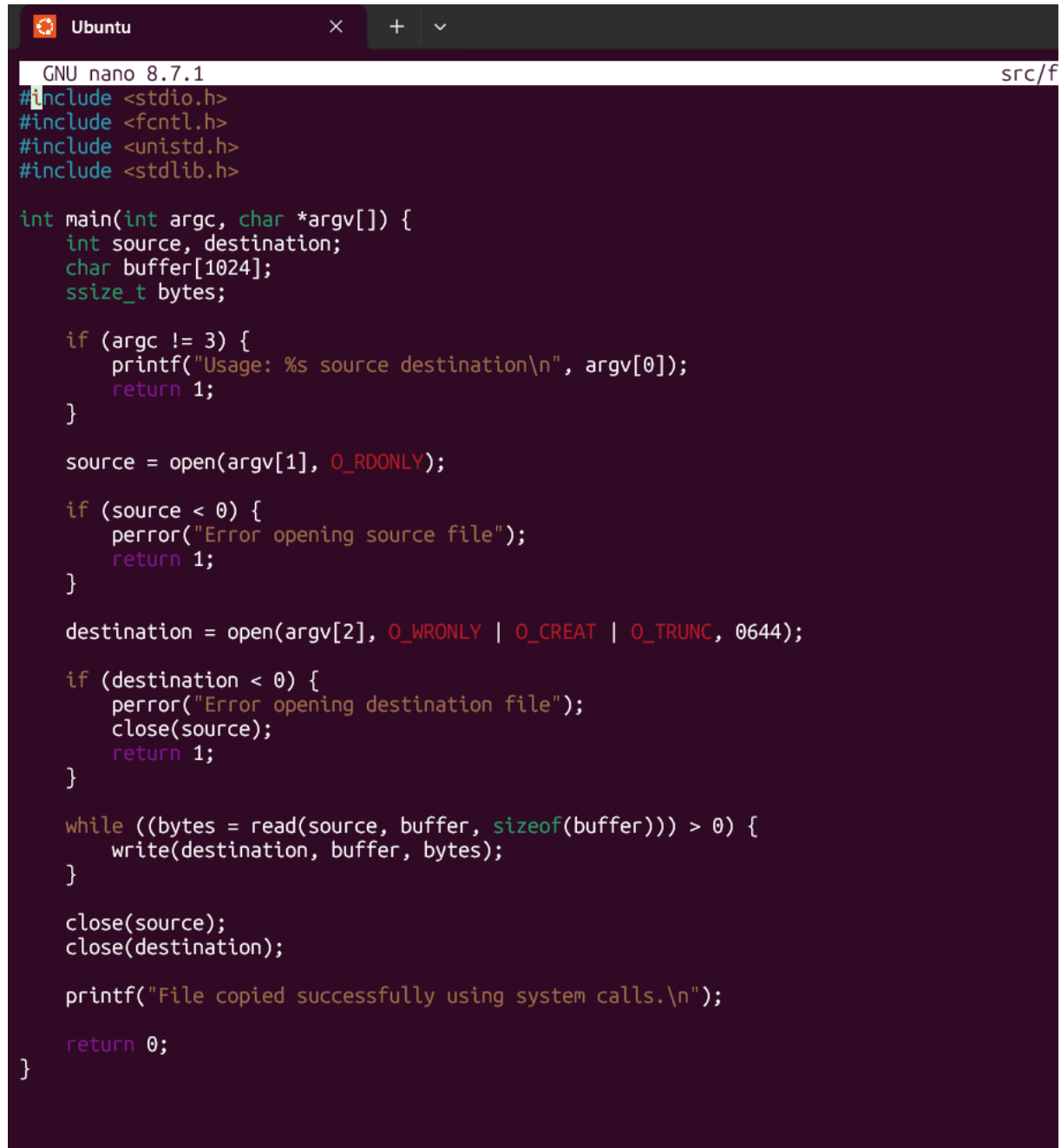


Practical Week 9

1. Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).



```
GNU nano 8.7.1 src/f
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main(int argc, char *argv[]) {
    int source, destination;
    char buffer[1024];
    ssize_t bytes;

    if (argc != 3) {
        printf("Usage: %s source destination\n", argv[0]);
        return 1;
    }

    source = open(argv[1], O_RDONLY);

    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }

    destination = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    while ((bytes = read(source, buffer, sizeof(buffer))) > 0) {
        write(destination, buffer, bytes);
    }

    close(source);
    close(destination);

    printf("File copied successfully using system calls.\n");

    return 0;
}
```

Output:

```
Ubuntu
nocturnal@nocturnal:~/OS_Skill/Week-9$ gcc src/file_copy_syscalls.c -o bin/file_copy_syscalls
nocturnal@nocturnal:~/OS_Skill/Week-9$ ./bin/file_copy_syscalls input.txt copy.txt
File copied successfully using system calls.
nocturnal@nocturnal:~/OS_Skill/Week-9$ cat copy.txt
Hello Sujit! This is a file copy test.
nocturnal@nocturnal:~/OS_Skill/Week-9$
```

Program to use lseek():

```
Ubuntu
GNU nano 8.7.1 src/file_copy
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main(int argc, char *argv[]) {
    int source, destination;
    char buffer[1024];
    ssize_t bytes;
    off_t size;

    if (argc != 3) {
        printf("Usage: %s source destination\n", argv[0]);
        return 1;
    }

    source = open(argv[1], O_RDONLY);

    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }

    destination = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    size = lseek(source, 0, SEEK_END);
    printf("Source file size: %ld bytes\n", (long)size);

    lseek(source, 0, SEEK_SET);

    while ((bytes = read(source, buffer, sizeof(buffer))) > 0) {
        write(destination, buffer, bytes);
    }

    close(source);
    close(destination);

    printf("File copied successfully using system calls.\n");

    return 0;
}
```

Output:

```
nocturnal@nocturnal:~/OS_Skill/Week-9$ nano src/file_copy_syscalls.c
nocturnal@nocturnal:~/OS_Skill/Week-9$ gcc src/file_copy_syscalls.c -o bin/file_copy_syscalls
nocturnal@nocturnal:~/OS_Skill/Week-9$ ./bin/file_copy_syscalls input.txt copy_lseek.txt
Source file size: 39 bytes
File copied successfully using system calls.
nocturnal@nocturnal:~/OS_Skill/Week-9$ cat copy_lseek.txt
Hello Sujit! This is a file copy test.
nocturnal@nocturnal:~/OS_Skill/Week-9$ D
```

2. Develop a program that redirects standard input and output using file descriptors and `dup2()`. Demonstrate how shells implement I/O redirection.

```
Ubuntu × + ▾
GNU nano 8.7.1 src/io_redirection
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int input_fd, output_fd;

    input_fd = open("input.txt", O_RDONLY);

    if (input_fd < 0) {
        perror("input.txt");
        return 1;
    }

    output_fd = open("output.txt",
                    O_WRONLY | O_CREAT | O_TRUNC,
                    0644);

    if (output_fd < 0) {
        perror("output.txt");
        close(input_fd);
        return 1;
    }

    dup2(input_fd, STDIN_FILENO);
    dup2(output_fd, STDOUT_FILENO);

    close(input_fd);
    close(output_fd);

    char buffer[100];

    printf("This output is redirected to output.txt\n");
    printf("Reading from redirected input:\n");

    while (read(STDIN_FILENO, buffer, sizeof(buffer) - 1) > 0) {
        break;
    }

    return 0;
}
```

Output:

```
Ubuntu × + ▾
nocturnal@nocturnal: ~/OS_Skill/Week-8$ nano src/io_redirection.c
nocturnal@nocturnal: ~/OS_Skill/Week-8$ echo "Hello Sujit! This is redirected input." > input.txt
nocturnal@nocturnal: ~/OS_Skill/Week-8$ cat input.txt
Hello Sujit! This is redirected input.
nocturnal@nocturnal: ~/OS_Skill/Week-8$ gcc src/io_redirection.c -o bin/io_redirection
nocturnal@nocturnal: ~/OS_Skill/Week-8$ ./bin/io_redirection
nocturnal@nocturnal: ~/OS_Skill/Week-8$ cat output.txt
This output is redirected to output.txt
Reading from redirected input:
nocturnal@nocturnal: ~/OS_Skill/Week-8$
```